ATOOSA AYAZBAKHSH ZIYABARI

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Master's student in Mechanical Engineering at the University of Alberta, with a B.Sc. in Electrical Engineering from K. N. Toosi University. Passionate about robotics and artificial intelligence, with strong technical and interpersonal skills and a proven ability to collaborate in teams and deliver projects successfully.

EDUCATION

University of Alberta — **Edmonton, AB, Canada**

Master of Science (M.Sc.) in Mechanical Engineering | *GPA: 3.9/4.0* — *2025 – Present*

- **Thesis:** Dynamic Difficulty Adjustment in Robotic Rehabilitation and Motor Learning
- **Relevant Coursework:** Assessment and Analysis of Biomechanical Motion (MEC E 686, A+), Machine Learning for Biomedical Applications (BME 677, A+), Applied Computational Intelligence for Engineers (MEC E 694, A+), Signal Processing of Time and Spectral Series (MEC E 653, A)
- **Honours:** University of Alberta Graduate Recruitment Scholarship (Jan 2025) – a competitive, merit-based scholarship awarded to incoming graduate students in recognition of academic excellence and research potential

K. N. Toosi University of Technology (K. N. T. U.) — **Tehran, Iran**

Bachelor of Science (B.Sc.) in Electrical Engineering | *GPA: 3.3/4.0* — *2017 – 2022*

- **Thesis:** Design of a digital blood pressure monitor and oscillometric signal processing

PUBLICATIONS

[1] **A. Ayazbakhsh Ziyabari**, R. Behbahani Nejad, A. Afkhami Ardekani, A. H. Vette, and M. Nazarahari, "Stay in the Zone: Intelligent Two-Layered Dynamic Difficulty Adjustment for Upper-Limb Exergame," *IEEE RAS/EMBS Int. Conf. on Biomedical Robotics and Biomechatronics (BioRob)*, 2026. DOI: 10.1109/BioRob66782.2026.11681391

[2] A. Afkhami Ardekani, A. Salemi, R. Behbahani Nejad, **A. Ayazbakhsh Ziyabari**, A. H. Vette, and M. Nazarahari, "Beyond the Score: Assessing Motor Learning via Inverse Reinforcement Learning from Suboptimal Human-in-the-Loop Gameplay," *IEEE RAS/EMBS Int. Conf. on Biomedical Robotics and Biomechatronics (BioRob)*, 2026. DOI: 10.1109/BioRob66782.2026.11681014

RESEARCH & WORK EXPERIENCE

University of Alberta — **Edmonton, AB, Canada**

Research Assistant, IDEA Lab – Dynamic Difficulty Adjustment in Robotic Rehabilitation and Motor Learning — *Jan 2025 – Present*

- Under the supervision of Dr. Milad Nazarahari and Prof. Albert H. Vette, designing control strategies and data-driven algorithms to optimize robotic rehabilitation systems.
- Focused on enhancing motor learning and personalizing therapy through adaptive difficulty adjustment and real-time performance analytics.

Teaching Assistant – MEC E 694: Applied Computational Intelligence for Engineers — *Sep 2026 – Present*

- Managed and lectured a 3-hour class session of the course; hold office hours and mark course assignments.

Teaching Assistant – MEC E 420: Feedback Control Design of Dynamic Systems (Prof. Albert H. Vette) — *Jan 2026 – Apr 2026*

- Served as lab instructor and teaching assistant for an undergraduate feedback control course, leading weekly lab sessions on the design and analysis of dynamic control systems.
- Guided students through MATLAB/Simulink modelling and controller-design exercises and held office hours to support student learning outcomes.

ATP Co. — **Tehran, Iran**

IT Manager — *Sep 2017 – Nov 2024*

- Oversaw and maintained the company's IT infrastructure, including systems and network administration, website management, and troubleshooting critical issues.
- Delivered reliable technical support and implemented solutions to ensure seamless operations.

Neuromatch Academy — **Remote**

Summer Intern – Exploring Memory and Storytelling: NLP Analysis of Narrative Types — *Jun 2023 – Aug 2023*

- Investigated the relationship between memory and storytelling by analysing cognitive processes such as imagination, recollection, and retelling using the Hippocorpus dataset of English narratives.
- Built and evaluated deep learning models in Python to identify patterns in how stories evolve under these conditions, covering data preprocessing, feature extraction, training, and performance assessment. Available as a Google Colab Notebook.

Zistel Company / FPGA Laboratory, K. N. T. U. — **Tehran, Iran**

Research Assistant – Blood Pressure Estimation Algorithm and Monitor Circuit Design — *Feb 2021 – May 2022*

- Under the supervision of Dr. Danial Katoozian and Dr. Hossein Hosseini-Nezhad, developed an algorithm for accurate blood pressure estimation.
- Compared Model-Based, Maximum Amplitude, Derivative Oscillometry, and Feedforward Neural Network algorithms.
- Designed a cost-efficient circuit to extract the oscillometric signal using a Sallen-Key active filter.

K. N. Toosi University of Technology — **Tehran, Iran**

Teaching Assistant – C Programming Course (M. Zamanian) — *Feb 2022 – May 2022*

- Assisted in teaching an undergraduate C programming course, guiding students through core concepts such as data types, control flow, functions, pointers, and memory management.

- Supported students during lab sessions and office hours and graded assignments and projects.

Teaching Assistant – Design of Digital Systems (FPGA, ASIC) Course (Dr. Hossein Hosseini-Nezhad) *Oct 2021 – May 2022*
- Assisted in teaching a digital system design course covering FPGA and ASIC design flows, guiding students through HDL (VHDL/Verilog) coding, simulation, synthesis, and timing analysis.
- Supported lab sessions and projects and evaluated student designs and assignments.

LEADERSHIP & VOLUNTEER EXPERIENCE

Faculty of Engineering Graduate Research Symposium (FEGRS) 2026, University of Alberta | *Volunteer* *Aug 2026*
- Recognized with a Certificate of Appreciation from the Associate Dean, Research Programs & Development.

Design Conference, University of Alberta | *Conference Volunteer* *Nov 2025*
- Assisted in organizing and facilitating conference sessions, ensuring smooth operations and participant engagement.
- Supported logistics, registration, and on-site coordination to enhance the overall event experience.

Biomedical Conference, University of Alberta | *Conference Volunteer* *May 2025*
- Supported the successful execution of a biomedical research conference by assisting with session coordination, participant engagement, and logistical operations.
- Ensured smooth registration and on-site organization and provided technical support to enhance the attendee experience.

SELECTED PROJECTS

Lung Cancer Image Classification with Deep Learning | *BME 677 Course Project, University of Alberta* *Fall 2025*
- Developed and compared PyTorch models (ResNet50, DenseNet121, and a CNN-Transformer hybrid) to classify lung images into normal, benign, and malignant classes, and ran an ablation study.
- Evaluated performance using ROC-AUC, F1/precision/recall, Cohen's kappa, and confusion matrices, with training dashboards and attention visualizations for model interpretability.

Affective State (Stress) Classification Using Physiological Signals (WESAD) | *MEC E 694 Course Project* *Fall 2025*
- Built an end-to-end Python/Jupyter pipeline for stress detection using the WESAD wearable dataset, including signal preprocessing, feature extraction across time/frequency/phase domains, and dimensionality reduction/representation learning (PCA, autoencoder features).
- Trained and compared global vs. personalized models and explored unsupervised clustering plus continuous stress/arousal estimation with temporal smoothing for more realistic monitoring.

SLAM and Boundary Detection Using 3D-LiDAR | *Autonomous Drive and Navigation (MEC E 652)* *Winter 2025*
- Built and evaluated a ROS (C++) 3D LiDAR SLAM + road-boundary/curb detection pipeline, combining LOAM/LeGO-LOAM feature-based odometry/mapping with road segmentation, RANSAC curve fitting, and Gaussian Process refinement.
- Validated on the KITTI dataset using RViz visualization and EVO trajectory/APE error analysis.

Bioinformatic Feature Analysis and Machine Learning | *Self-Directed Project* *Spring 2023*
- Used Jupyter Notebook to code in Python, finding features associated with the "type" and selecting the top 50 features most highly associated with the "Tumour" or "Normal" label.
- Trained a logistic regression model and predicted a label for each sample in the test file. Available on GitHub.

PPG-Based Biosensing: PCB Design + Blood Glucose Estimation | *Zistel Company, K. N. T. U.* *Summer 2022*
- Under the supervision of Dr. Danial Katoozian, designed and fabricated a PPG signal acquisition PCB in Altium Designer and developed a classic ML/feature-based pipeline to estimate blood glucose from PPG signals.
- Focused on signal preprocessing and vital feature extraction, then evaluated candidate features/algorithms under limited-data constraints to improve estimation accuracy.

Space Invaders Game | *Advanced Programming Course, K. N. T. U.* *Winter 2022*
- Wrote the game in Java using an object-oriented approach, with encapsulation, polymorphism, composition, nested classes, and the JavaFX library. Available on GitHub.

CERTIFICATIONS

- **Reinforcement Learning Specialization** | University of Alberta – Coursera — Jul 2025
- **Deep Learning – Tutorial and Project Certificate** | Neuromatch Academy — Jul 2023
- **IELTS – Academic Module** | CEFR Level: C1 | Band Score: 7 (S: 7, W: 6.5, R: 7.5, L: 7.5) — Apr 2023
- **Neural Networks and Deep Learning** | DeepLearning.AI – Coursera | Grade: 95.62/100 — Mar 2022
- **Python Data Structures** | University of Michigan – Coursera | Grade: 93.6/100 — Mar 2022
- **Machine Learning** | Stanford University – Coursera | Grade: 93.21/100 — Jan 2022

SKILLS & LANGUAGES

Programming Languages: MATLAB, Python, C, C++, C#, Java, VHDL, Assembly
Machine Learning & AI: PyTorch, scikit-learn, Deep Learning (CNNs, Transformers, Autoencoders), Computer Vision, NLP, Reinforcement Learning, LLMs
Other Skills: ROS (Robot Operating System), PCB Design, Analogue and Digital Signal Processing, ARM (nRF), Git, Graphic Design (Photoshop, Illustrator), Windows Server, WordPress, Power BI, Tableau, Python visualization libraries
Languages: Persian (Native), English (Advanced), French (Basic)